# MACHINE LEARNING

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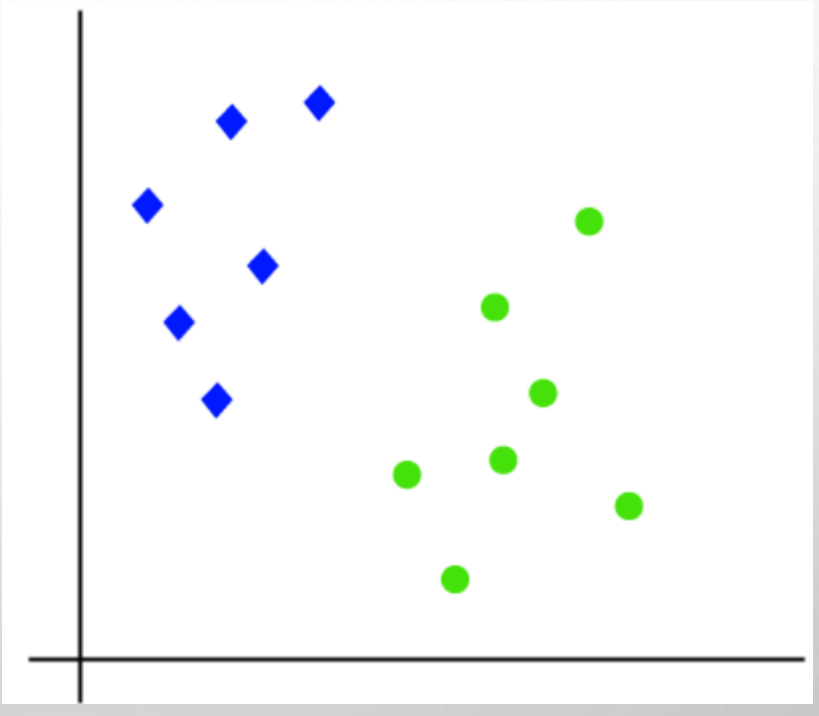
# LECTURE 4

Classification (part 3)

SUPPORT VECTOR MACHINES (SVM)

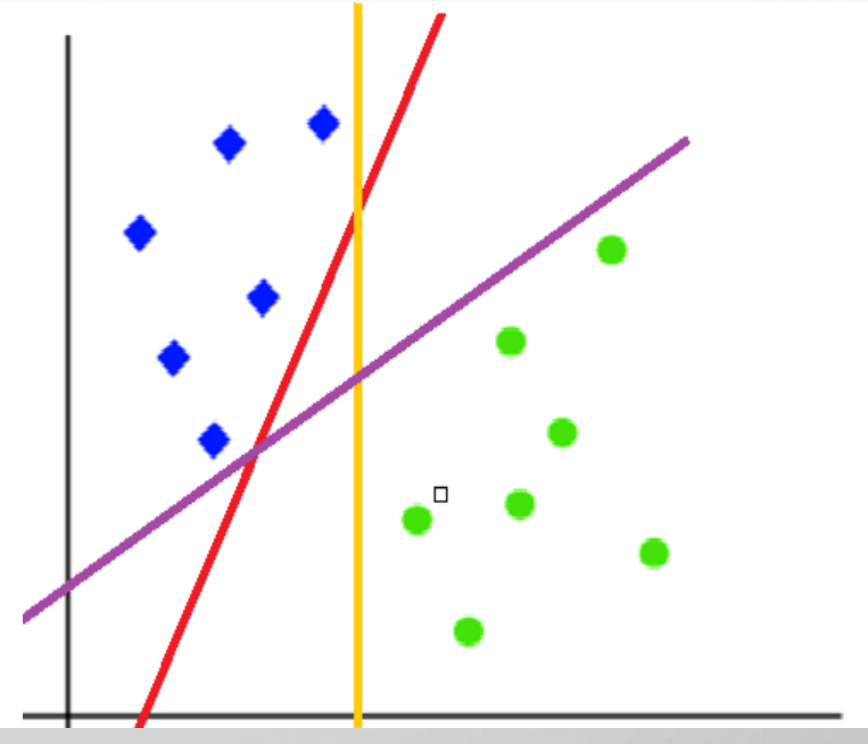
# FROM LOGISTIC REGRESSION TO SVM

As we know, we need to find the best **“hyperplane”.** For 2D plot it will be a **straight line** but if we have more dimensions, we call this decision boundary.



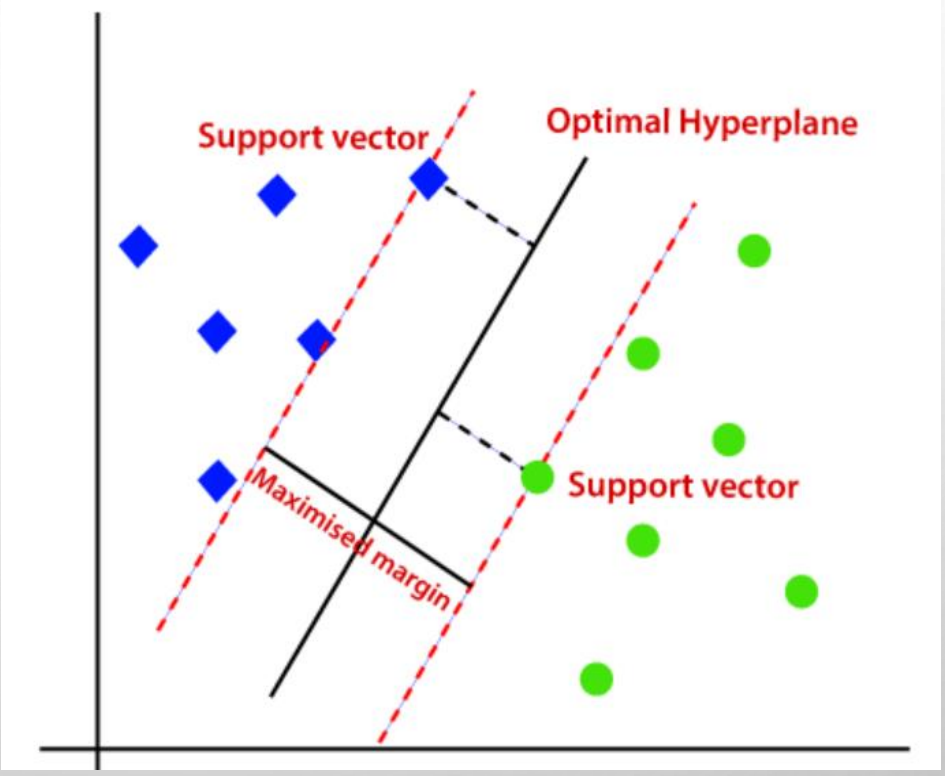
# FROM LOGISTIC REGRESSION TO SVM

For logistic regression, the classifier is defined over all the points.



# FROM LOGISTIC REGRESSION TO SVM

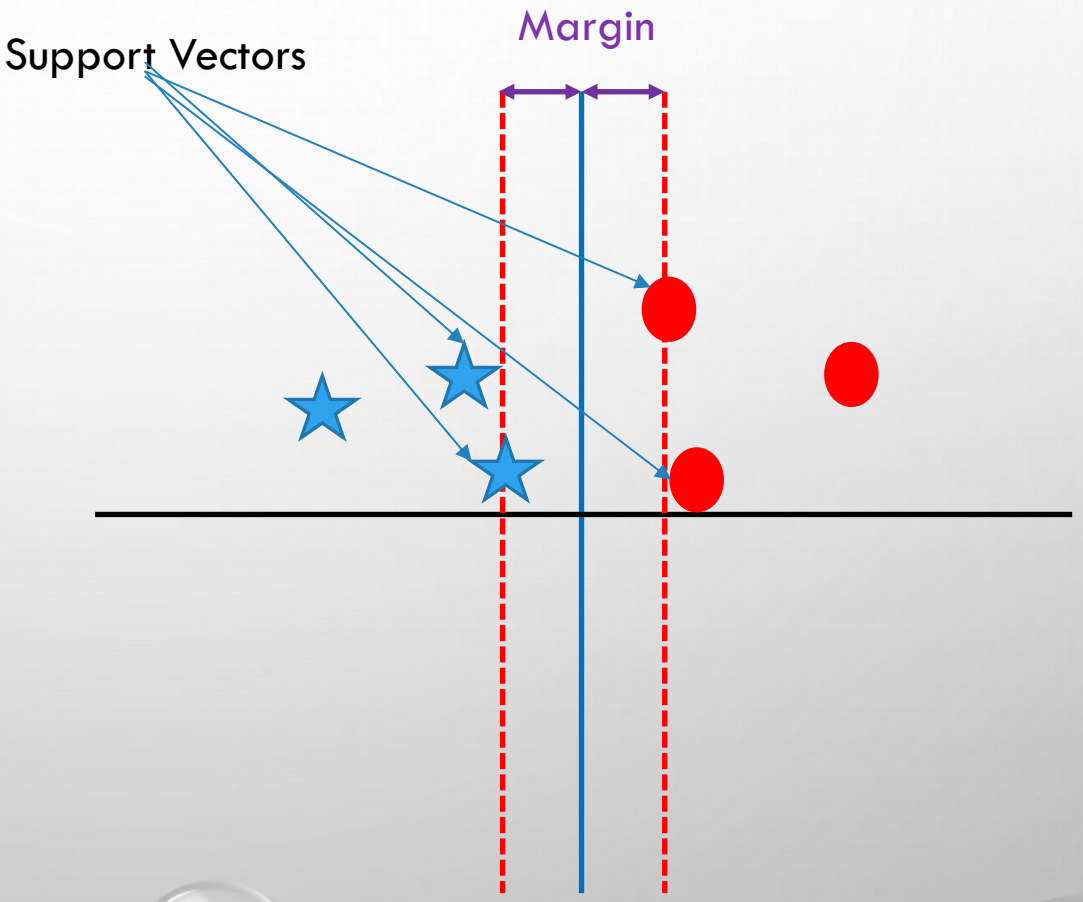
The best hyperplane is the plane that has the maximum distance from both classes, and this is the main aim of SVM. This is done by finding the hyperplane that maximize the margin ( maximize the distance between the hyperplane and the datapoints closed to it ( support vectors).



# LARGE MARGIN CLASSIFIER (SVM)

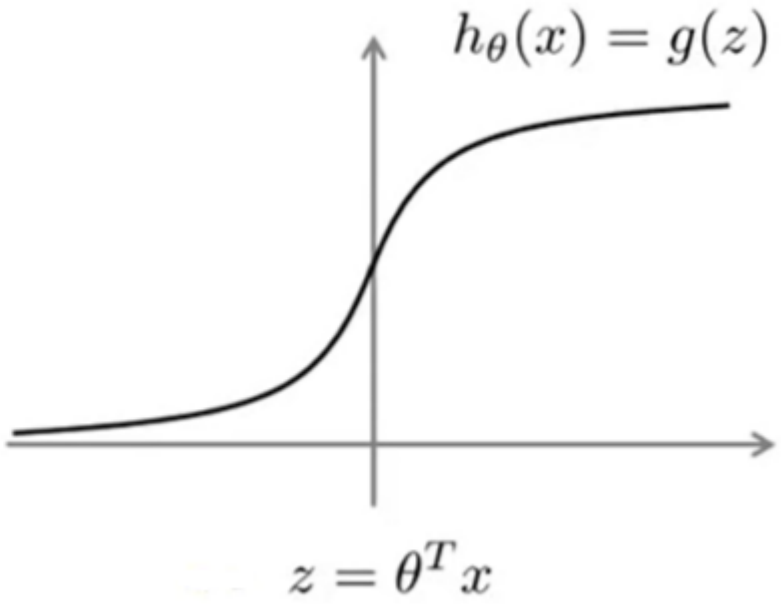
Support Vectors :

- They are the datapoints that define the maximum margin for the hyperplane for the dataset.
- They are difficult points to classify.
- The hyperplane is sensitive to their positions (any change will affect the formation of hyper plane).



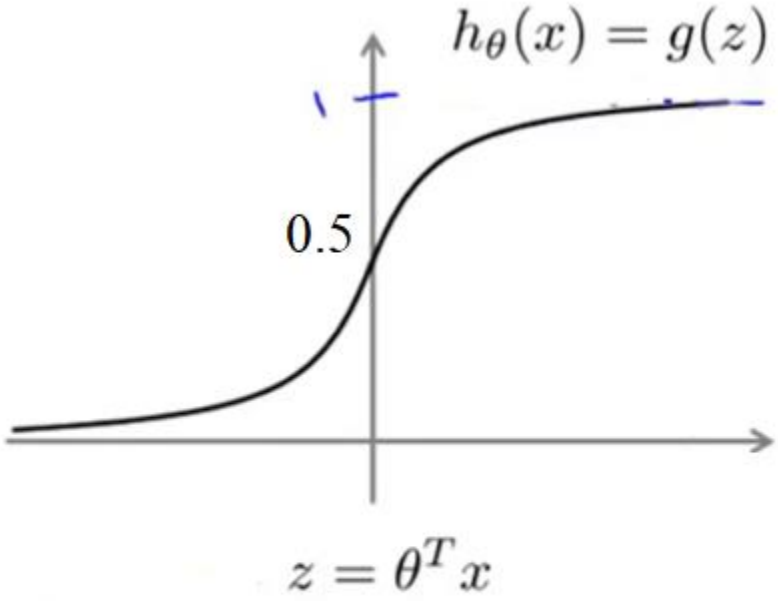
# FROM LOGISTIC REGRESSION TO SVM

$$h_\theta(x) = \frac{1}{1 + e^{-\theta^T x}}$$



# FROM LOGISTIC REGRESSION TO SVM

$$h_\theta(x) = \frac{1}{1 + e^{-\theta^T x}}$$




If $y = 1$, we want $h_\theta(x) \approx 1$, $\theta^T x \gg 0$

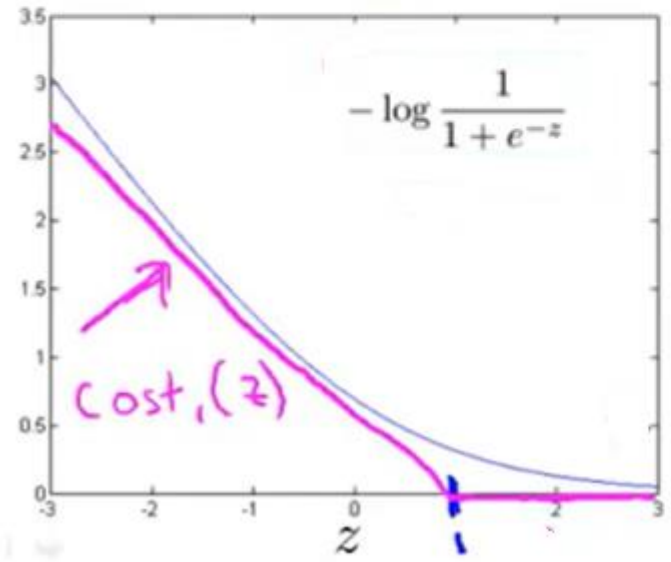
If $y = 0$, we want $h_\theta(x) \approx 0$, $\theta^T x \ll 0$

# FROM LOGISTIC REGRESSION TO SVM

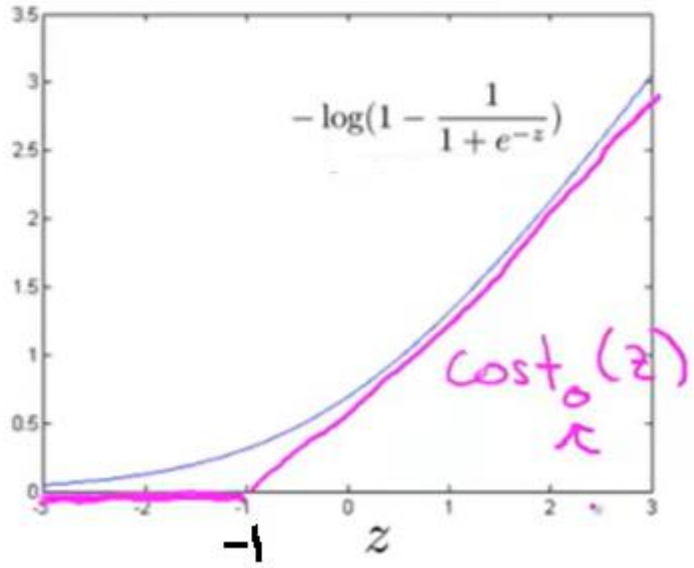
Cost of example: $-(y \log h_\theta(x) + (1-y)\log(1-h_\theta(x)))$

$$= -\,\mathrm{y} \log \frac{1}{1+e^{-\theta^T x}} - (1-\mathrm{y}) \log(1 - \frac{1}{1+e^{-\theta^T x}})$$

If $y = 1$ (want $\theta^T x \gg 0$):




If $y = 0$ (want $\theta^T x \ll 0$):



**Logistic regression:**

$$\min_{\theta} \frac{1}{m}\left[\sum_{i=1}^{m} y^{(i)} \underbrace{\left(-\log h_\theta(x^{(i)})\right)}_{cost_1(\theta^T x^{(i)})} + (1-y^{(i)}) \underbrace{\left((-\log(1-h_\theta(x^{(i)}))\right)}_{cost_0(\theta^T x^{(i)})}\right] + \frac{\lambda}{2m}\sum_{j=1}^{n}\theta_j^2$$

**Support vector machine:**

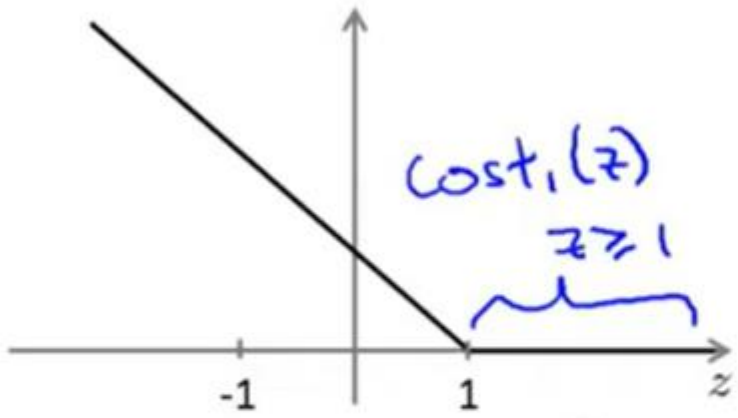
$$\min \frac{1}{m} \quad [\sum_{i=1}^{m} y^{(i)} cost_1(\theta^T x^{(i)}) + (1-y^{(i)}) cost_0(\theta^T x^{(i)})] + \frac{\lambda}{2m}\sum_{j=1}^{n}\theta_j{}^2$$

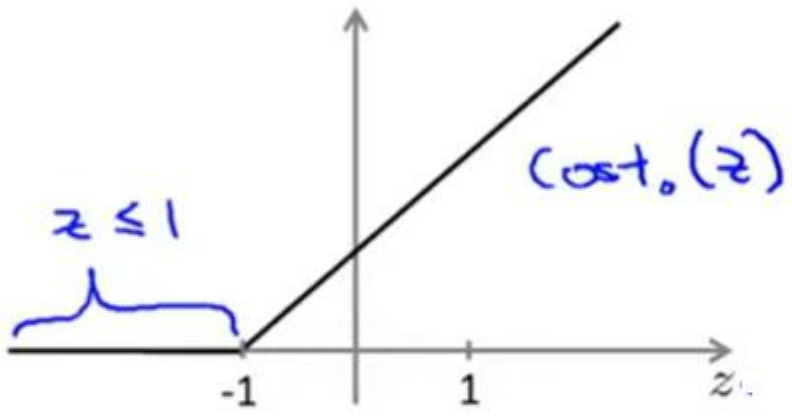
By removing 1/m, and multiply by 1/λ
Let C = 1/λ

$$\min_{\theta} C\sum_{i=1}^{m}\left[y^{(i)} cost_1(\theta^T x^{(i)}) + (1-y^{(i)}) cost_0(\theta^T x^{(i)})\right] + \frac{1}{2}\sum_{j=1}^{n}\theta_j^2$$

# FROM LOGISTIC REGRESSION TO SVM

$$\min_{\theta} C \sum_{i=1}^{m} \left[ y^{(i)} cost_1(\theta^T x^{(i)}) + (1 - y^{(i)}) cost_0(\theta^T x^{(i)}) \right] + \frac{1}{2} \sum_{i=1}^{n} \theta_j^2$$

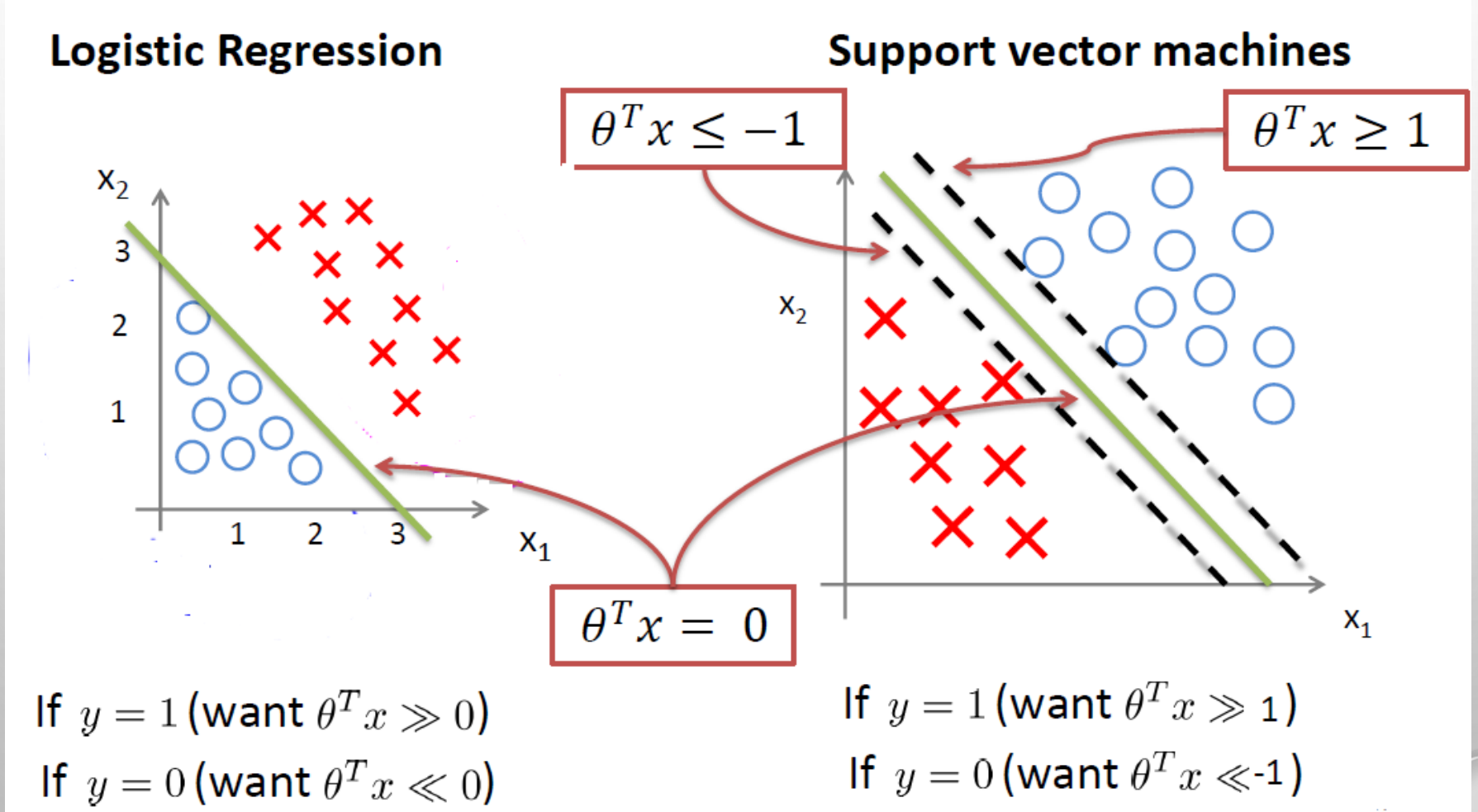



If $y = 1$, we want $\theta^T x \geq 1$ (not just $\geq 0$)

If $y = 0$, we want $\theta^T x \leq -1$ (not just $< 0$)

# DIFFERENCES BETWEEN LOGISTIC REGRESSION AND SVM DECISION BOUNDARY



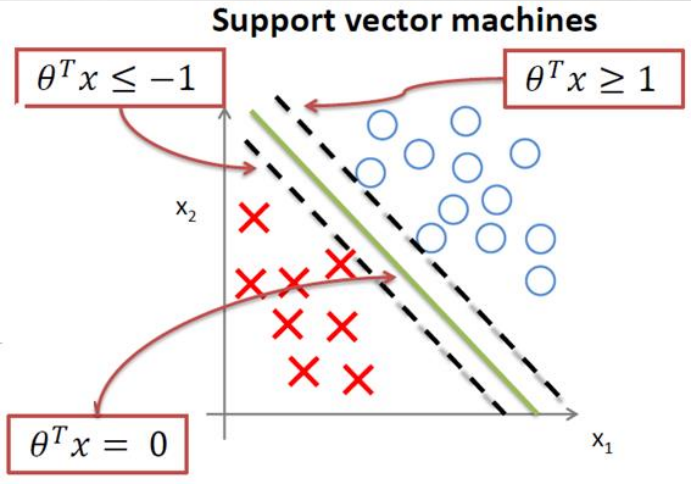
# SVM OPTIMIZATION PROBLEM

$$\min_{\theta} C \sum_{i=1}^{m} \left[ y^{(i)} cost_1(\theta^T x^{(i)}) + (1 - y^{(i)}) cost_0(\theta^T x^{(i)}) \right] + \frac{1}{2} \sum_{j=1}^{n} \theta_j^2$$

Classification Error

Margin Error

$$\text{s.t.} \quad \theta^T x^{(i)} \geq 1 \qquad \text{if } y^{(i)} = 1$$
$$\theta^T x^{(i)} \leq -1 \qquad \text{if } y^{(i)} = 0$$



# SVM OPTIMIZATION PROBLEM

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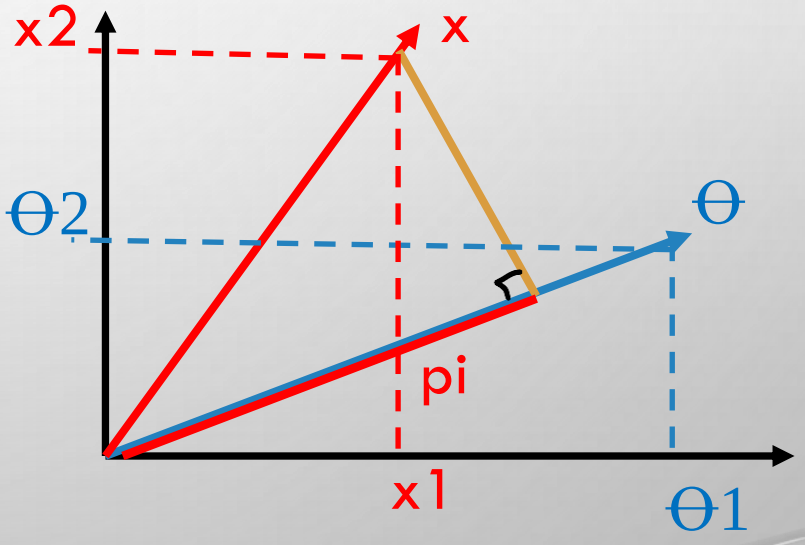
Classification Error

Margin Error

$$\text{s.t.} \quad \theta^T x^{(i)} \geq 1 \qquad \text{if } y^{(i)} = 1$$
$$\theta^T x^{(i)} \leq -1 \qquad \text{if } y^{(i)} = 0$$

$$\theta^T \mathrm{x} = p^{(i)} \, \|\theta\|$$



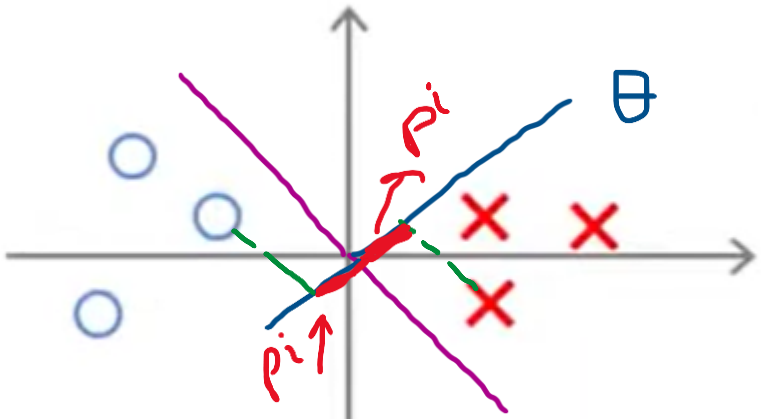
# SVM OPTIMIZATION PROBLEM

$$\text{s.t.} \quad p^{(i)} \cdot \|\theta\| \geq 1 \qquad \text{if } y^{(i)} = 1$$

$$p^{(i)} \cdot \|\theta\| \leq -1 \qquad \text{if } y^{(i)} = 1$$

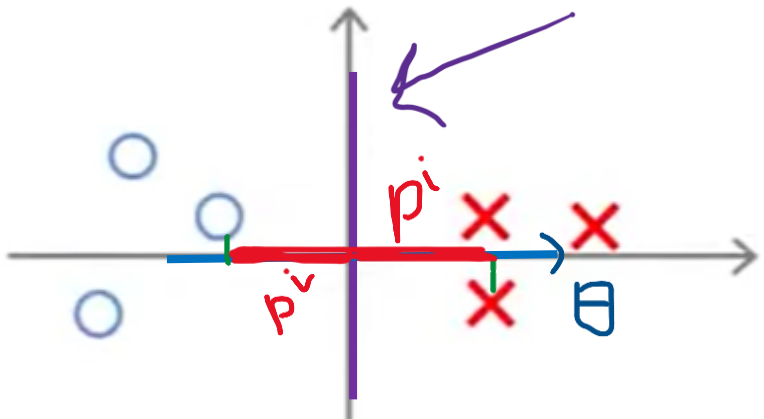
where $p^{(i)}$ is the projection of $x^{(i)}$ onto the vector $\theta$.

Simplification: $\theta_0 = 0$




P (i) is too small , which means Θ will be large X

Chosen boundary by SVM

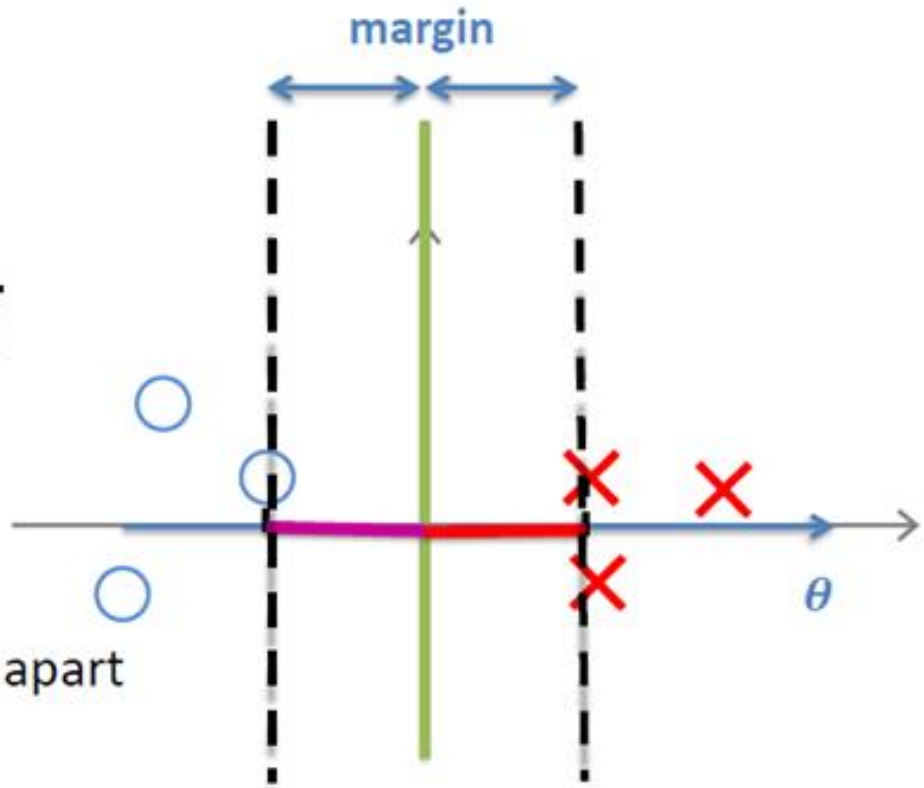



P (i) is large , which means Θ will be small √

# MARGIN CALCULATION

$$\text{Margin} = \frac{|\theta^T x|}{\|\theta\|} + \frac{|\theta^T x|}{\|\theta\|} = \frac{1}{\|\theta\|} + \frac{1}{\|\theta\|} = \frac{2}{\|\theta\|}$$

$$= \frac{2}{\sqrt{\theta_1{}^2 + \theta_2{}^2}}$$




**Margin error:**

measure of how the two dashed lines are far apart

If lines are close → large margin error

If lines are far apart → small margin error

Margin error = $\theta_1{}^2 + \theta_2{}^2$

Regularization Term

# PARAMETER C (PENALTY)

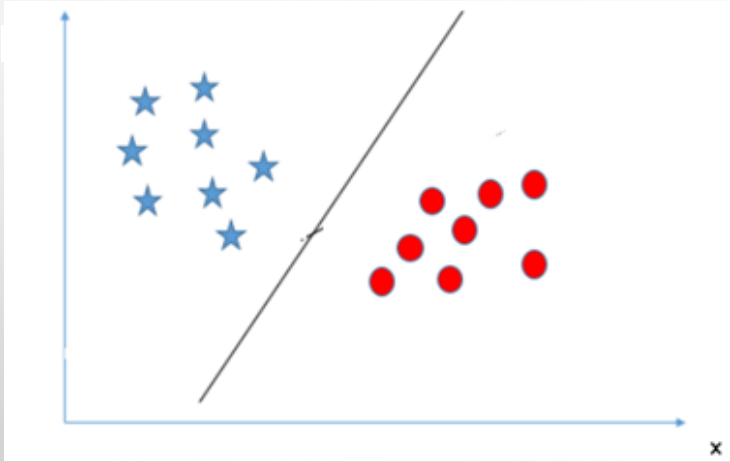
$$\min_{\theta} C \sum_{i=1}^{m} \left[ y^{(i)} cost_1(\theta^T x^{(i)}) + (1 - y^{(i)}) cost_0(\theta^T x^{(i)}) \right] + \frac{1}{2} \sum_{j=1}^{n} \theta_j^2$$

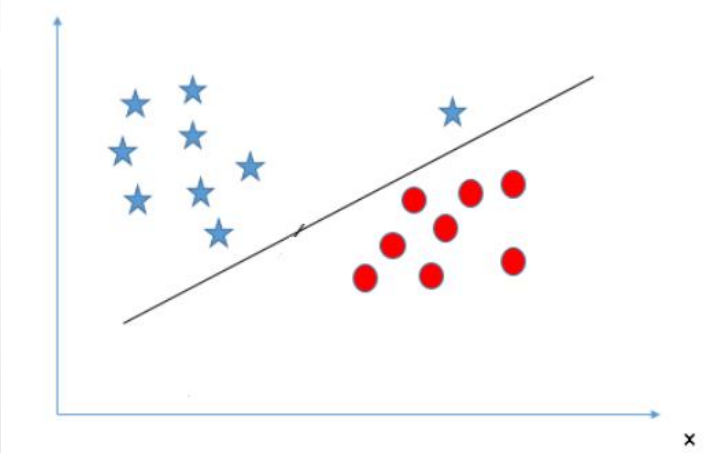
Classification Error

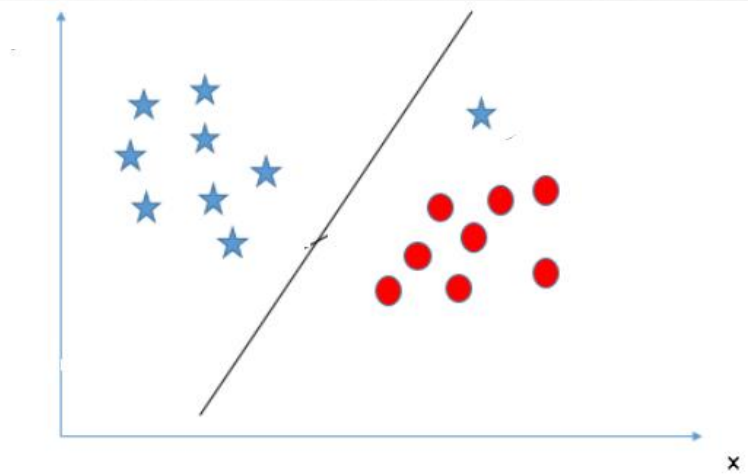
Margin Error

The best value for C ??

# PARAMETER C (PENALTY)




The value of C will choose which line to consider

Large C -> hard margin
Small C -> soft margin

# PARAMETER C (PENALTY)

Very Large C means Classification error tend to zero. Thus, we reach hard margin that support even outliers. However, poor generalization.

$$\min_{\theta} C \sum_{i=1}^{m} \left[ y^{(i)} cost_1(\theta^T x^{(i)}) + (1 - y^{(i)}) cost_0(\theta^T x^{(i)}) \right] + \cancel{\frac{1}{2} \sum_{j=1}^{n} \theta_j^2}$$

Classification Error

Margin Error

Small C means we reach soft margin that leads to better generalization.

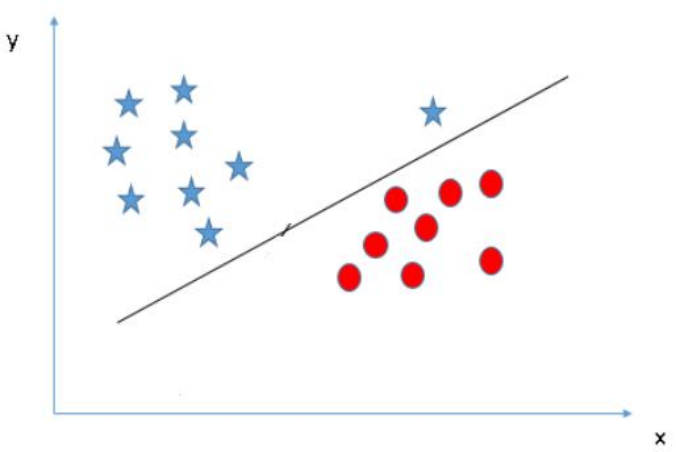
$$\min_{\theta} C \sum_{i=1}^{m} \cancel{\left[ y^{(i)} cost_1(\theta^T x^{(i)}) + (1 - y^{(i)}) cost_0(\theta^T x^{(i)}) \right]} + \frac{1}{2} \sum_{j=1}^{n} \theta_j^2$$

Classification Error

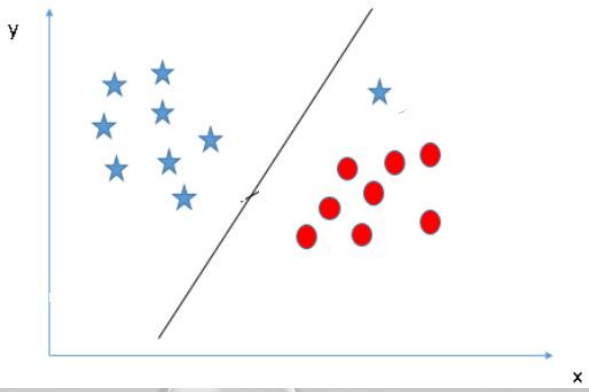
Margin Error

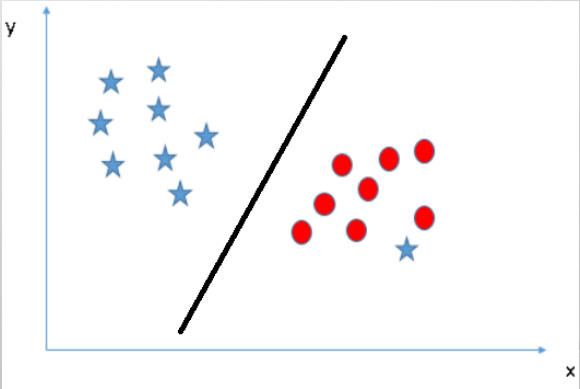
# PARAMETER C (PENALTY)

C is large (lambda is small) will cause overfitting (hard margin).




C is small (lambda is large) will cause better generalization (soft margin).



# SVM OPTIMIZATION PROBLEM

Optimization problem:

$$\min_{\theta} C \sum_{i=1}^{m} \left[ y^{(i)} cost_1(\theta^T x^{(i)}) + (1 - y^{(i)}) cost_0(\theta^T x^{(i)}) \right] + \frac{1}{2} \sum_{j=1}^{n} \theta_j^2$$

$$\text{s.t.} \quad \theta^T x^{(i)} \geq 1 \qquad \text{if } y^{(i)} = 1$$

$$\theta^T x^{(i)} \leq -1 \qquad \text{if } y^{(i)} = 0$$

Optimization problem solution:

$$\boxed{\theta = \sum_{i=1}^{m} \alpha_i x_i} \qquad \boxed{\theta^T x_j = \sum_{i=1}^{m} \alpha_i x_i^{\ T} . x_j}$$

# KERNEL MATRIX

Example : if we have three training examples

$$X^{(1)} = \begin{pmatrix} x_1^1 \\ x_2^1 \end{pmatrix}, X^{(2)} = \begin{pmatrix} x_1^2 \\ x_2^2 \end{pmatrix} \text{ and } X^{(3)} = \begin{pmatrix} x_1^3 \\ x_2^3 \end{pmatrix} \qquad y^{(1)} = class-,\ y^{(2)} = class+ \ and\, y^{(3)} = class+$$

We need to find $\theta_0 + \theta_1\, x_1 \quad + \theta_2\, x_2$

Take bias into consideration

$$X^{(1)} = \begin{pmatrix} 1 \\ x_1^1 \\ x_2^1 \end{pmatrix}, X^{(2)} = \begin{pmatrix} 1 \\ x_1^2 \\ x_2^2 \end{pmatrix} \text{and } X^{(3)} = \begin{pmatrix} 1 \\ x_1^3 \\ x_2^3 \end{pmatrix}$$

Kernel matrix will be equal

$$\begin{bmatrix} X^{(1)T}X^{(1)} & X^{(2)T}X^{(1)} & X^{(3)T}X^{(1)} \\ X^{(1)T}X^{(2)} & X^{(2)T}X^{(2)} & X^{(3)T}X^{(2)} \\ X^{(1)T}X^{(3)} & X^{(2)T}X^{(3)} & X^{(3)T}X^{(3)} \end{bmatrix} \qquad X^{(i)T}X^{(j)} = 1 + x_1^i\, x_1^j + x_2^i\, x_2^j$$

# KERNEL MATRIX

Kernel matrix

$$\begin{bmatrix} X^{(1)T}X^{(1)} & X^{(2)T}X^{(1)} & X^{(3)T}X^{(1)} \\ X^{(1)T}X^{(2)} & X^{(2)T}X^{(2)} & X^{(3)T}X^{(2)} \\ X^{(1)T}X^{(3)} & X^{(2)T}X^{(3)} & X^{(3)T}X^{(3)} \end{bmatrix} \begin{bmatrix} \alpha_1 \\ \alpha_2 \\ \alpha_3 \end{bmatrix}$$

$$\propto_1 X^{(1)T}X^{(1)} + \propto_2 X^{(2)T}X^{(1)} + \propto_3 X^{(3)T}X^{(1)} = -1$$
$$\propto_1 X^{(1)T}X^{(2)} + \propto_2 X^{(2)T}X^{(2)} + \propto_3 X^{(3)T}X^{(2)} = 1$$
$$\propto_1 X^{(1)T}X^{(3)} + \propto_2 X^{(2)T}X^{(3)} + \propto_3 X^{(3)T}X^{(3)} = 1$$

Solve for $\propto_1, \propto_2$ and $\propto_3$ then substitute

$$\theta = \begin{pmatrix} \theta_0 \\ \theta_1 \\ \theta_2 \end{pmatrix} = \propto_1 \begin{pmatrix} 1 \\ x_1^1 \\ x_2^1 \end{pmatrix} + \propto_2 \begin{pmatrix} 1 \\ x_1^2 \\ x_2^2 \end{pmatrix} + \propto_3 \begin{pmatrix} 1 \\ x_1^3 \\ x_2^3 \end{pmatrix}$$

$The\ boundry\ is\ ready$ $\theta_0 + \theta_1 x_1^t + \theta_2 x_2^t$ t is the index of testing example

# SVM OPTIMIZATION PROBLEM

$$\theta^T x_j = \sum_{i=1}^{m} \alpha_i \boxed{x_i^{\ T} . x_j}$$

Kernel matrix

Which means that we can express the inner-product of θ with any input $x_j$ purely in terms of inner-products between training inputs

During test time, to make a prediction on test input $x_t$ , we use a classifier in terms of inner-products between the test point and the training points

$$\text{s.t.} \quad \theta^T x^{(i)} \geq 1 \qquad \text{if } y^{(i)} = 1$$
$$\theta^T x^{(i)} \leq -1 \qquad \text{if } y^{(i)} = 0$$

If sign> 0, classify as positive

$$\underline{classifier} = \underline{sign(\textstyle\sum_{i=1}^{m} \alpha_i x_i^{\ T} . x_t)}$$

If sign<0, classify as negative

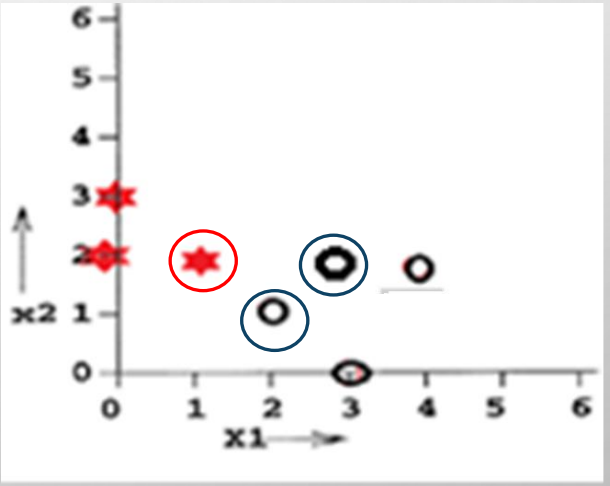
# SVM EXAMPLE

Regarding linear SVM, given a dataset with two features x1 and x2 and target y as shown in the figure. The positive points are shown as stars and the negative points are circles.

1) find the support vectors from the graph on the right
2) Calculate the decision boundry parameters using Lagrange multipliers
3) Given a test case X = (4, 1), what will be the SVM response.

S1 = (1, 2) S2 = (2, 1) S3= (3, 2)

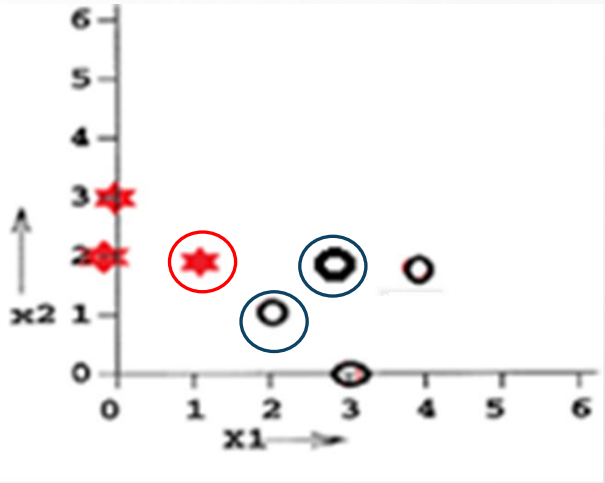
S1' = (1, 2, 1) S2' = (2, 1, 1) S3' = (3, 2, 1)



# SVM EXAMPLE

2)

S1' = (1, 2, 1)  S2' = (2, 1, 1)  S3' = (3, 2, 1)




a1 S1'.S1' +a2 S2'.S1' + a3 S3'.S1' = 1
a1 S1'.S2' +a2 S2'.S2' + a3 S3'.S2' = -1
a1 S1'.S3' +a2 S2'.S3' + a3 S3'.S3' = -1

6a1 + 5a2 + 8 a3= 1
5a1 + 6a2 + 9 a3 = -1
8a1 + 9a2 + 14 a3 = -1

a1 =1  a2= -1 a3= 0

$\Theta = (1,2,1)^T - (2,1,1)^T = (-1,1,0)^T$

Decision line = -x1 +x2

# SVM EXAMPLE

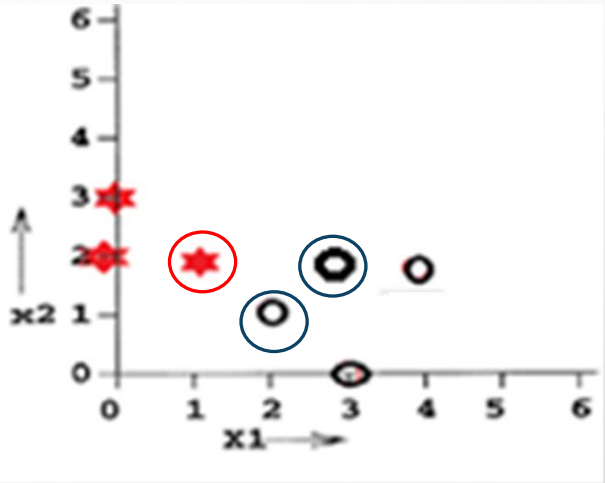
3)

Decision line = -x1 +x2

Test case X = ( 4, 1)

By substitution in decision boundry
- 4 + 1 = -3 => -ve result

Thus, the test case is classified as negative class



# CREDIT FOR

- ***MACHINE LEARNING, BY ANDREW NG –UNI. OF STANFORD.***

# THANK YOU ☺